

Abigail J. Lee

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EDUCATION & APPOINTMENTS

University of California, Berkeley	Aug 2025 – Present
Miller Research Fellow (Aug 2026 – Jul 2029)	
NASA Hubble Fellow (Aug 2025 – Jul 2026)	
University of Chicago	May 2025
Ph.D. in Astronomy & Astrophysics	
Certificate in College Teaching	
Advised by Wendy Freedman	
University of Pennsylvania	May 2019
B.A. in Physics <i>summa cum laude</i> with Departmental Honors	
Minors in Classical Studies and Mathematics	
Advised by Gary Bernstein	

AWARDS & FELLOWSHIPS

IAU Ph.D. Prize, Division J (Galaxies and Cosmology)	2026
Miller Research Fellowship	2025
NASA Hubble Fellowship	2025
CCAPP Price Prize in Cosmology and AstroParticle Physics	2024
AGB Stars Prize for Young Researchers, XIV Torino Workshop on AGB Stars	2024
Future Investigators in NASA Earth and Space Science and Technology (FINESST)	2022
Illinois Space Grant Consortium Graduate Fellowship	2022
LSST Cooperation Data Science Fellowship	2021
Chambliss Astronomy Achievement Award	2021
Elaine K. Bernstein Women in Science Award, UChicago	2019
McCormick Graduate Fellowship, UChicago	2019
NASA Pennsylvania Space Grant Undergraduate Scholarship	2018
University Scholar, Penn	2015

PREVIOUS RESEARCH POSITIONS

Graduate Research Assistant, UChicago	2019 – 2025
Summer Visiting Researcher, Seoul National University	2023
Undergraduate Research Assistant (Cosmology), Penn	2017 – 2019
Summer Undergraduate Research Assistant, Stanford/KIPAC	2018
Summer Undergraduate Researcher, Max Planck Institute for Gravitational Physics	2017
Summer Intern, NASA Jet Propulsion Laboratory	2017
Undergraduate Research Assistant (Experimental Condensed Matter), Penn	2016 – 2017

OBSERVING PROPOSALS & GRANTS

Summary: Awarded \$1.4 million in competitive funding as PI

As Principal Investigator/Submitter

James Webb Space Telescope

- “J-Virgo: A JWST Treasury Survey of the Virgo Cluster”, co-PI with Daniel Weisz, 147 hours (NIRCam) / 67 hours (NIRISS), \$673,498, 2025

Hubble Space Telescope

- “Quantifying Systematics in the JAGB Method Distance Scale in M33”, Archival, \$19,678, 2023
- “Quantifying Systematics in the JAGB Method Distance Scale in M31”, Archival, \$30,886, 2022

Full Stipendiary Grants

- NASA Hubble Fellowship, “AGB Stars in the Era of NIR Astronomy: New Probes of Cosmology and Galaxy Evolution”, \$148,059, 2025-2026
- Miller Research Fellowship, “AGB Stars in the Era of IR Astronomy: New Probes of Cosmology and Galaxy Evolution”, \$354,000, 2026-2029
- NASA FINESST, “Measuring the Hubble Constant with Carbon Stars”, \$150,000, 2022-2025

Keck (DEIMOS)

- “Investigating a potential dwarf galaxy accretion event with stellar kinematics”, 1 night, 2026B

Magellan-Baade Telescope (FourStar)

- “JAGB and TRGB Distances to Three New Galaxies”, 3 nights, 2024B

As Co-Investigator (select programs)

Roman Space Telescope GIP-2002 (PI: K. Gilbert), 2026

Euclid GIP24-0005 (PI: A. Savino), 2025

James Webb Space Telescope GO-1995 (PI: W. Freedman), 2021

UChicago Midway Research II Computing Allocation (PI: J. Frieman), 123k hours, 2021

Hubble Space Telescope AR-16126 (PI: W. Freedman), 2020

Hubble Space Telescope AR-16127 (PI: W. Freedman), 2020

PUBLICATIONS

FIRST-AUTHOR PUBLICATIONS

1. **Lee, A. J.**, Weisz, D. R., Dolphin, A., Savino, A. *The Star Formation History of WLM from Asymptotic Giant Branch Stars and The Discovery of a Possible Accreted System in its Outer Disk*, 2026, [arXiv:2603.00243](#), submitted to ApJ
2. **Lee, A. J.**, Weisz, D. R., Ren, Y., Savino, A., Dolphin, A. *Measuring Star Formation Histories from Asymptotic Giant Branch Stars I: A Demonstration in M31*, 2025, [ApJ](#), **995**, 135
3. **Lee, A. J.**, Freedman, W. L., Madore, B. F., Jang, I. S., Owens, K. A., Hoyt, T. J. *The Chicago-Carnegie Hubble Program: The JWST J-region Asymptotic Giant Branch (JAGB) Extragalactic Distance Scale*, 2025, [ApJ](#), **985**, 182

4. **Lee, A. J.**, Monson, A. J., Freedman, W. L., Madore, B. F., Owens, K. A., Beaton, R. L., Espinoza, C., Ren, T., Ren, Y. *Resolved Near-infrared Stellar Photometry from the Magellan Telescope for 13 Nearby Galaxies: JAGB Method Distances*, 2024, [ApJ, 967, 22](#)
5. **Lee, A. J.**, Freedman, W. L., Jang, I. S., Madore, B. F., Owens, K. A. *First JWST Observations of JAGB Stars in the SN Ia host galaxies: NGC 7250, NGC 4536, NGC 3972*, 2024, [ApJ, 961, 132](#)
6. **Lee, A. J.** *Carbon Stars as Standard Candles: An Empirical Test for the Metallicity, Age, and Reddening Sensitivity of the J-region Asymptotic Giant Branch (JAGB) Method*, 2023, [ApJ, 956, 15](#)
7. **Lee, A. J.**, Rousseau-Nepton, L., Freedman, W. L., Madore, B. F., Cioni, M.-R., Hoyt, T. J., Jang, I. S., Javadi, A., Owens, K. A. *The Astrophysical Distance Scale: V. A 2% Distance to the Local Group Spiral M33 via the JAGB Method, Tip of the Red Giant Branch, and Leavitt Law*, 2022, [ApJ, 933, 201](#)
8. **Lee, A. J.**, Freedman, W. L., Madore, B. F., Owens, K. A., Jang, I. S. *A Preliminary Calibration of the JAGB Method Using Gaia EDR3*, 2021, [ApJ, 923, 157](#)
9. **Lee, A. J.**, Freedman, W. L., Madore, B. F., Owens, K. A., Monson, A. J., Hoyt, T. J. *The Astrophysical Distance Scale III: Distance to the Local Group Galaxy WLM using Multi-Wavelength Observations of the Tip of the Red Giant Branch, Cepheids, and JAGB Stars*. 2021, [ApJ, 907, 112](#)

CO-AUTHOR PUBLICATIONS

1. Sanderson, R. E., ..., **Lee, A.**, ..., *A near field guide to Roman's wide-area surveys*, 2026, [arXiv:2603.12981](#), submitted to AAS Journals
2. Owens, K. A., Jang, I. S., Freedman, W. L., Hoyt, T. J., **Lee, A. J.**, Madore, B. F., Monson, A. J. *Investigating the Extragalactic Distance Scale with JWST: Cepheids in NGC 7250*, 2026, submitted to ApJ
3. Hoyt, T. J., Jang, I. S., Freedman, W. L., Madore, B. F., Owens, K. A., **Lee, A. J.** *The Chicago Carnegie Hubble Program: Improving the Calibration of SNe Ia with JWST Measurements of the Tip of the Red Giant Branch*, 2026, [ApJ, 1002, 11](#)
4. Freedman, W. L., Madore, B. F., Jang, I. S., Hoyt, T. J., **Lee, A. J.**, Owens, K. A. *Status Report on the Chicago-Carnegie Hubble Program (CCHP): Measurement of the Hubble Constant Using the Hubble and James Webb Space Telescopes*, 2025, [ApJ, 985, 203](#)
5. Freedman, W. L., Madore, B. F., Hoyt, T. J., Jang, I. S., **Lee, A. J.**, Owens, K. A. *I-Band Asymptotic Giant Branch (IAGB) Stars: II. A First Estimate of their Precision and a Differential Zero Point*, 2025, [AJ, 169, 247](#)
6. Madore, B. F., Freedman, W. L., Hoyt, T. J., Jang, I. S., **Lee, A. J.**, Owens, K. A. *I-Band Asymptotic Giant Branch (IAGB) Stars: I. Exploring a New Standard Candle for the Extragalactic Distance Scale*, 2025, [AJ, 169, 162](#)

7. Hoyt, T. J., Jang, I. S., Freedman, W. L., Madore, B. F., **Lee, A. J.**, Owens, K. A. *Coordinated JWST Imaging of Three Distance Indicators in a SN Host Galaxy and an Estimate of the TRGB Color Dependence*, 2024, [ApJ](#), **975**, 111
8. Porth, L., Bernstein, G., Smith, R. E., **Lee, A. J.** *The Information Content of Projected Galaxy Fields*, 2022, [MNRAS](#), **518**, 3344
9. Madore, B. F., Freedman, W. L., **Lee, A. J.**, Owens, K. A. *Milky Way Zero-Point Calibration of the JAGB Method: Carbon Stars in Galactic Open Clusters*, 2022, [ApJ](#), **938**, 125
10. Owens, K. A., Freedman, W. L., Madore, B. F., **Lee, A. J.** *Current Challenges in Cepheid Distance Calibrations Using Gaia EDR3*, 2022, [ApJ](#), **927**, 8
11. Madore, B. F., Freedman, W. L., **Lee, A. J.** *Astrophysical Distance Scale IV. Preliminary Zero-Point Calibration of the JAGB Method in the HST/WFC3-IR Broad J-Band (F110W) Filter*. 2022, [ApJ](#), **926**, 153
12. Vishnubhotla, R., Ping. J., Gao, Z., **Lee, A.**, Saouaf, O., Vrudhula, A., Johnson, A. T. C. *Scalable Graphene Aptasensors for Drug Quantification* 2017, [AIP Advances](#) **7**, 115111

NON-REFEREED MANUSCRIPTS

1. Kraemer, K., et al. including **Lee, A. J.**, *Setting the Stage for Improving the Distance Ladder with Roman Core Community Surveys*, 2023, [White Paper](#) submitted to the Roman Core Community Survey
2. **Lee, A. J.**, et al. *Predicting Winners of the Reality TV Dating Show The Bachelor Using Machine Learning Algorithms*, 2022, [April Fools arXiv paper](#)

INVITED COLLOQUIA & SEMINARS

UC Davis Astrophysics & Cosmology seminar	Dec 2025
Florida State University Astrophysics seminar	Apr 2025
Washington University in St. Louis Physics Theory seminar	Jan 2025
LSST Dark Energy Science Collaboration (DESC) seminar	Sep 2024
The Ohio State University CCAPP Price Prize seminar	Aug 2024
Seoul National University Astronomy seminar	Sep 2023
University of Pennsylvania Astrophysics seminar	Oct 2022
Argonne National Lab Cosmology seminar	Oct 2021

CONFERENCE TALKS & POSTERS

*Talk The stellar route to H_0 : not only Classical Cepheids, Switzerland (Invited)	Jul 2026
Talk NHFP Symposium, Baltimore	Oct 2025
Plenary Observational Tensions in Cosmology, Naples (Invited)	May 2025
Dissertation Talk AAS 245, National Harbor	Jan 2025
Plenary CERN EuCAPT online Rapid Response Workshop on H_0 (Invited)	Sep 2024
Talk XIV Torino Workshop on AGB stars, Rome	Jun 2024
Poster Science with HST and JWST VII, Porto	Apr 2024
Talk AAS 243, New Orleans	Jan 2024

Talk Giant Branch Distance Scale Workshop, Baltimore (Invited)	Nov 2023
Talk MIAPbP workshop on the extragalactic distance scale, Munich (Invited)	Jul 2023
Talk AAS 241, Seattle	Jan 2023
Talk AAS 240, Pasadena	Jun 2022
Poster AAS 237 (Winner of Chambliss Award)	Jan 2021
Talk Gulf Coast Undergraduate Research Symposium, Houston	Nov 2017
Talk Emerging Researchers National Conference in STEM, D.C.	Mar 2017

*Upcoming

RESEARCH ADVISING

Coral Espinoza , Lake Forest College (LFC) undergraduate Oversaw UChicago REU summer project and LFC senior thesis	2022 – 2023
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TEACHING

Guest Lecturer , UChicago, <i>Physics of Stars</i>	2023
Guest Lecturer , UChicago, <i>Galaxies</i>	2020
Teaching Assistant , UChicago, <i>Galaxies</i>	2020
Teaching Assistant , UChicago, <i>Stars</i>	2019
Teaching Assistant , Penn, <i>Classical Mechanics</i>	2019
Observing Labs Teaching Assistant , Penn, <i>Survey of the Universe</i>	2018, 2019
Teaching Assistant , Penn, <i>E&M, Optics, and Modern Physics</i>	2018
Lab Teaching Assistant , Penn, <i>Classical Mechanics</i>	2017
Tutor , Penn Physics Department	2016, 2017

OBSERVING

Public data – significant experience with data from JWST, HST, Gaia, 2MASS, VMC

Magellan/FourStar Imaging - 20 nights

Magellan/IMACS Imaging - 1 night

Magellan/IMACS Spectroscopy - 1 night

COLLABORATION MEMBERSHIP

Dark Energy Spectroscopic Instrument (DESI)	2026 –
<i>Roman</i> Science Collaboration, Nearby Galaxies Working Group	2025 –
Chicago-Carnegie Hubble Program (CCHP)	2019 –

PRESS

Expertise Interviews:

- PopSci, [Is the universe really infinite?](#)
- PopSci, [JWST takes a jab at the mystery of the universe's expansion rate](#)

Selected press featuring [Lee et al. 2025a](#):

- Quanta, [The Webb Telescope Further Deepens the Biggest Controversy in Cosmology](#)

- Dr. Becky YouTube Channel, [JWST turned the Crisis in Cosmology into a bigger problem](#)
- Dr. Becky YouTube Channel, [Has JWST SOLVED the crisis in cosmology?!](#)

Press featuring [Lee et al. 2021a](#):

- UChicago News, [Aging stars provide a new cosmological yardstick](#)

SELECTED OUTREACH AND EDI

Youtube “Cosmology Talks” “Extragalactic distances from the JAGB method”	2024
Public Talk “The Expansion of the Universe”, Lake Forest Astronomy Club	2022
Public Talk “Mentorship in Academia”, UChicago Physics of Stars highschool class	2021
Bay Area Scientists Inspiring Students (BASIS)	2025 – Present
AMP-UP Mentor for 2 PhD students	2025 – 2026
Astrobites, my articles (Ombudsperson 23-25, Hiring Committee 23-24)	2022 – 2025
Inclusion, Diversity and Equity in Astronomy, UChicago, (Organizer 22-24)	2019 – 2025
UChicago Empowerment of North Koreans (ENOK), English Tutor	2022 – 2023
Advising & Mentoring Program, UChicago, Organizer	2020 – 2023
Peer Mentor for 3 PhD Students, UChicago	2020 – 2023
Chambliss Award Judge	2022
Space Explorers Winter Institute Instructor, UChicago	2020
Observing Nights Organizer, Penn	2018 – 2019
Penn Veterans Upward Bound, Astronomy Tutor for Veterans	2018 – 2019

SERVICE & DEPARTMENT COMMITTEES

Referee for ApJ, ApJL, A&A (4+ papers)	
Reviewer for NASA FINESST and international funding proposals	
Miller Institute Annual Symposium, Organizing Committee	2026 – Present
Galaxy Formation (GalForm) Group, Organizer, Berkeley	2025 – Present
Graduate Admissions Committee, UChicago	2024 – 2025
Climate Survey Committee, Student Representative, UChicago	2021 – 2024
Faculty Meeting Graduate Student Representative, UChicago	2019 – 2020
Society of Physics Students, Social Chair, Penn	2018 – 2019

OTHER ACTIVITIES

Languages: English (native), Spanish (intermediate), Korean (basic)	
Rec Soccer, play in several high-level (college experience) leagues 3x/week	2019 – Present
Semi-professional Women’s Soccer (WPSL, WMLIS)	2023 – 2025
Penn Women’s Club Soccer Team	2015 – 2019
Penn Symphony Orchestra, first violin	2015 – 2019